Biometric-based exam gate authentication system with SMS confirmation using Blockchain

A Mini Project Report Submitted
In partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology
in
Information Technology

by

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CERTIFICATE

This is to certify that this is the bonafide record of the mini project entitled **"Biometric-based exam gate authentication system with SMS confirmation using blockchain"**, submitted by **S.Greeshma(18N31A12E5), U.Bhargavi(18N31A12G8), Saiyog Nagari(18N31A12E2), S. Jeevan Sai Chowdary(18N31A12E0)** of B.Tech in the partial fulfillment of the requirements for the degree of Bachelor of Technology in Information Technology during the year 2020-2021. The results embodied in this mini project report have not been submitted to any other university or institute for the award of any Degree or Diploma.

Internal Guide

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DECLARATION

We hereby declare that the mini project titled "Biometric-based exam gate authentication system with SMS confirmation using blockchain" submitted to Malla Reddy College of Engineering and Technology(UGC Autonomous), affiliated to Jawaharlal Nehru Technological University Hyderabad (JNTUH) for the award of the degree of Bachelor of Technology in Information Technology is a result of original work carried-out in this project. It is further declared that the mini project report or any part thereof has not been previously submitted to any University or Institute for the award of Degree or Diploma.

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ABSTRACT

Distributed Ledged Technologies (DLT) commonly known as Blockchain is a thriving technology these days that is helping transform outmoded operations to faster, secured, and pocket-friendly approaches. It is a specific type of database that collects information together in groups, also known as blocks, that hold sets of information. Blocks have certain storage capacities and, when filled, are chained onto the previously filled block, forming a chain of data known as the “blockchain”. Blocks are formed in linear and chronological order and are immutable. Each block in the chain is given an exact timestamp when it is added to the chain. The potential of blockchain for biometrics is quite fascinating and mutually benefit each other. Biometrics use psychological traits such as the face, fingerprints, iris, voice, hand geometry, behavior characteristics helping to solve the security authentication puzzle.

To make biometrics work with blockchain, primarily we must create the fingerprint template, since the raw biometric data makes it costly for the user to store it in the block chain. Later after encryption using AES(Advanced Encryption Standard) the Cypher Text is stored using IPFS(InterPlanatery File System) which is later hashed and stored in the block chain. When the information is added to the blocks, each block will have its time stamp and contains its hash along with the hash of the before block. The hash function is a mathematical function that turns information into a string of letters and numbers which provides the required security to the information in the node. Indisputably blockchains provide greater security to biometrics, halting hackers to steal or modify the biometric information.

When this process meets another level of security called OTP (One time Password) verification, there will be no question of students or others indulging in fraudulent activities. Sending OTP through SMS (Small Message Services) uses GMS (Global System for Mobile Communications) telecommunications standard. When a student gets near the examination gate, an official is allowed to authenticate the student using a user interface and the biometrics which are already being recorded during the exam registration process. When the student biometrics is found to be matched, he/she has to authenticate using OTP. If biometrics are not matched after 3 attempts, he/she is not allowed into the examination center. Though biometrics are used by many institutions these days, our approach is for increasing its level of security and thereby decreasing spurious activities. This can be possible by adopting blockchain and OTP confirmation leading a path to 2-factor authentication.

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1 INTRODUCTION

1.1 Introduction

Blockchain in recent years has become an interesting aspect to be used in many real-life applications. Blockchain restores trust by putting it in the hands of many instead of relying on a centralized entity, which gives it the potential to become one of the most important breakthroughs of our time. This trust can then be used to share and store legal, medical, financial, or personal information. It is a distributed ledger database on a peer-to-peer (P2P) network that comprises a list of ordered blocks chronologically. Trust relation among distributed nodes is established by mathematical methods and cryptography technologies instead of semi-trusted central institutions.

The usage of biometrics with blockchain shows impressive results in maintaining security and trust. Biometrics include face recognition, voice recognition, fingerprint recognition, iris recognition, etc which act as basic information to be inserted as nodes into the blockchain. As biometric technology continue to advance, it is used in many areas of everyday life. They are used in airport security, law authentication, mobile access and authentication, banking, home assistants, building access, public transport, schools. Talking about schools, usage of biometrics is confined only to the attendance system. But one must recognize its usage in terms of security. It can be used in the exam entry authentication process which helps in the reduction of fraudulent activities and saves time.

Though there is the blockchain that provides ample security, usage of the OTP verification process could provide another shield of security. This can be done using GSM (Global System for Mobile communications) telecommunication standard developed by the European Telecommunications Standards Institute (ETSI) to describe the protocols for second-generation (2G) digital cellular networks used by mobile devices such as mobile phones and tablets. 2G networks developed as a replacement for first-generation (1G) analog cellular networks. The GSM standard originally described a digital, circuit-switched network optimized for full-duplex voice telephony. This expanded over time to include data communications, first by circuit-switched transport, then by packet data transport via General Packet Radio Service (GPRS), and Enhanced Data Rates for GSM Evolution (EDGE). Subsequently, the 3GPP developed third-generation (3G) UMTS standards, followed by fourth-generation (4G) LTE Advanced standards, which do not form part of the ETSI GSM standard. So this standard can be used with any kind of mobiles.

The main concept is providing security in the examination process, which is provided by the combination of blockchain, biometrics, and OTP authentication process.

Introduction to blockchain

A blockchain is a distributed software network that functions both as a digital ledger and a mechanism enabling the secure transfer of assets without an intermediary. It is sometimes referred to as Distributed Ledger Technology (DLT), which makes the history of any digital asset unalterable and transparent through the use of decentralization and cryptographic hashing. Just as the internet is a technology that facilitates the digital flow of information, blockchain is a technology that facilitates the digital exchange of units of value. Anything from currencies to land titles to votes can be tokenized, stored, and exchanged on a blockchain network. Thefirst manifestation of blockchain technology emerged in 2009 with the Bitcoin blockchain, a secure, censorship-resistant, peer-to-peer electronic cash system. Because Bitcoin is accessible to anyone, it is an example of an open or a permissionless blockchain. Today, there are many forms of blockchain technology. Some blockchains have been designed to meet the needs of a finite group of participants, where access to thenetwork is restricted. These are examples of private, or permission blockchains. Blockchain is an especially promising and revolutionary technology because it helps reduce risk, stamps out fraud, and brings transparency in a scalable way for myriad uses. Blockchain consists of 3 important concepts: blocks, nodes, miners.

Blocks:

Every chain consists of multiple blocks and each block has three basic elements:

- The **data** in the block.
- A 32-bit whole number called a **nonce.** The nonce is randomly generated when a block is created, which then generates a block header hash.
- The **hash** is a 256-bit number wedded to the nonce. It must start with a huge number of zeroes (i.e., beextremely small).

When the first block of a chain is created, a nonce generates the cryptographic hash. The data in the block isconsidered signed and forever tied to the nonce and hash unless it is mined.

Miners:

Miners create new blocks on the chain through a process called mining. In a blockchain every block has its unique nonce and hash, but also references the hash of the previous block in the chain, so mining a block isn't easy, especially on large chains. Miners use special software to solve the incredibly complex math problem of finding a nonce that generates an accepted hash. Because the nonce is only 32 bits and the hash

is 256, there are roughly four billion possible nonce-hash combinations that must be mined before the right one is found. When that happens miners are said to have found the "golden nonce" and their block is added to the chain. Making a change to any block earlier in the chain requires re-mining not just the block with the change, but all of the blocks that come after. This is why it's extremely difficult to manipulate the blockchain.

Nodes:

Nodes can be any kind of electronic device that maintains copies of the blockchain and keeps the network functioning. Every node has its copy of the blockchain and the network must algorithmically approve any newly mined block for the chain to be updated, trusted, and verified. Since blockchains are transparent, every action in the ledger can be easily checked and viewed. Each participant is given a unique alpha numeric identification number that shows their transactions. Combining public information with a system of checks and balances helps the blockchain maintain integrity and creates trust among users. Essentially, blockchains can be thought of as the scalability of trust via technology.

Differences between databases and blockchain:

Type	**Block Chain**	**Data Bases**
Authority	Blockchain is decentralized and has no centralized approach. However, there are private blockchains that may utilize some forms of centralization	Databases are controlled by the administrator and are centralized in nature.
Architecture	Blockchain uses a distributed ledger network architecture.	Database utilize a client-server architecture.
Data Handling	Blockchain utilizes Read and Write operations.	The databases support CRUD (Create, Read, Update and Delete).
Integrity	Blockchain supports integrity.	Malicious actors can alter database data.
Transparency	Public blockchain offers transparency.	Databases cannot be transparent. Only the administrator can decide which the public can access data.

Cost	Blockchains are comparatively harder to implement and maintain.	The database being an old technology is easy to implement and maintain.
Performance	Blockchain is bobbed down by the verification and consensus methods.	Databases are extremely fast and offer great scalability.
Use Cases	Storing confidential information.	Transfer value.

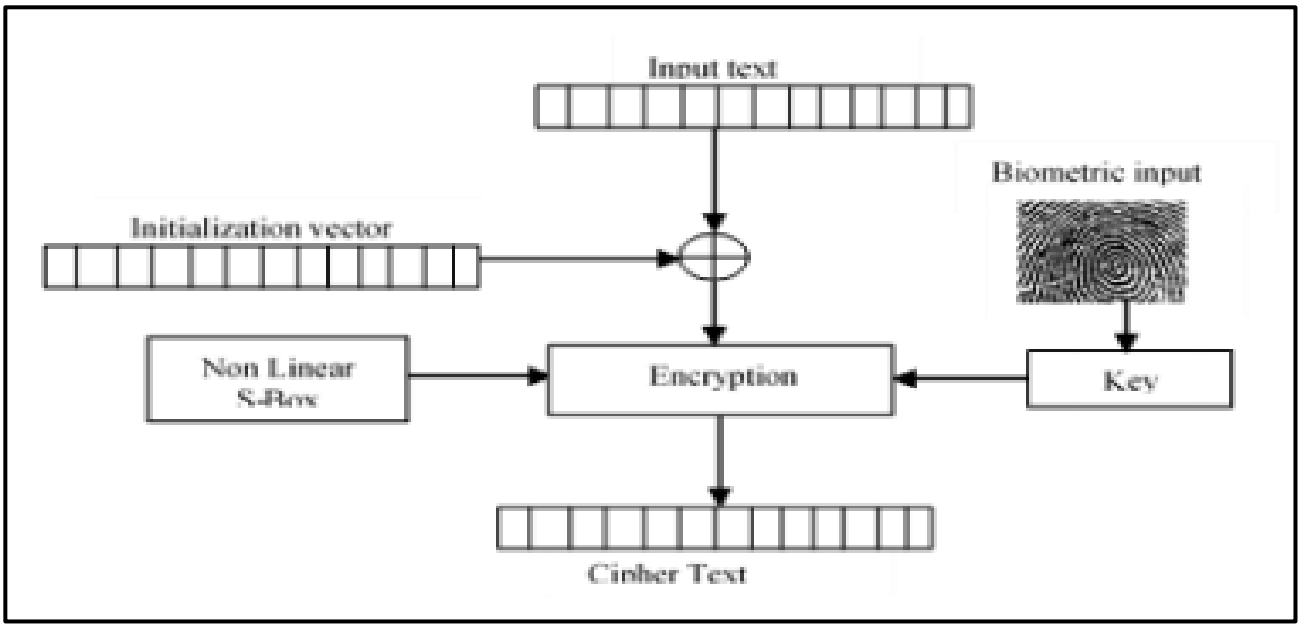
Algorithms used to secure information in the blockchain:

Biometrics, with its uniqueness to every individual, has been adapted as a security authentication feature by many institutions. These biometric data are processed into templates that are saved on databases, and a centralauthority centralizes and controls these databases. This form of storing biometric data, or in our case fingerprint template, is asymmetric and prone to three main security attacks, such as fake template input, template modification or deletion, and channel interception by a malicious attacker. To avoid this we have many algorithms used in blockchain technology. The fingerprint is encrypted by the symmetric key algorithm: Advanced Encryption Standard (AES) algorithm and then is uploaded to an asymmetrically distributed storagesystem, the InterPlanetary File system (IPFS). The hash of the templated is stored in a decentralized blockchain. The hash of the templated is stored in a decentralized blockchain. The slow transaction speed of the blockchain has limited its use in real-life applications, such as large file storage, hence, the merge with IPFS to store just the hashes of large files. The encrypted template is uploaded to the IPFS, and its returned digest is stored on the Ethereum network. The implementation of IPFS prevents storing the raw state of the fingerprint template on the Ethereum network to reduce cost and also prevent identity theft. This procedure isan improvement of previous systems. By adopting the method of template hashing, the proposed system is cost-effective and efficient.

- **Advanced Encryption Standard:**

  The more popular and widely adopted symmetric encryption algorithm likely to be encountered nowadaysis the Advanced Encryption Standard (AES). It is found at least six times faster than the triple-DES (Data Encryption Standard). AES is an iterative rather than a Feistel cipher. It is based on 'substitution– permutation network'. It comprises a series of linked operations, some of which involve replacing inputs with specific outputs (substitutions) and others involve shuffling bits around (permutations). AES performs all its computations on bytes rather than bits. Hence, AES treats the 128 bits of a plaintext blockas 16 bytes. These 16 bytes are arranged in four columns and four rows for processing as a matrix.

Encryption Process:



Here, we restrict to a description of a typical round of AES encryption. Each round comprises four sub-process.

Byte Substitution (SubBytes):

The 16 input bytes are substituted by looking up a fixed table (S-box) given in design. The result is in a matrix of four rows and four columns.

Shift rows:

Each of the four rows of the matrix is shifted to the left. Any entries that 'fall off' are re-inserted on the right side of the row. The shift is carried out as follows –

- The first row is not shifted.
- The second row is shifted one (byte) position to the left.
- The third row is shifted two positions to the left.
- The fourth row is shifted three positions to the left.
- The result is a new matrix consisting of the same 16 bytes but shifted concerning each other.

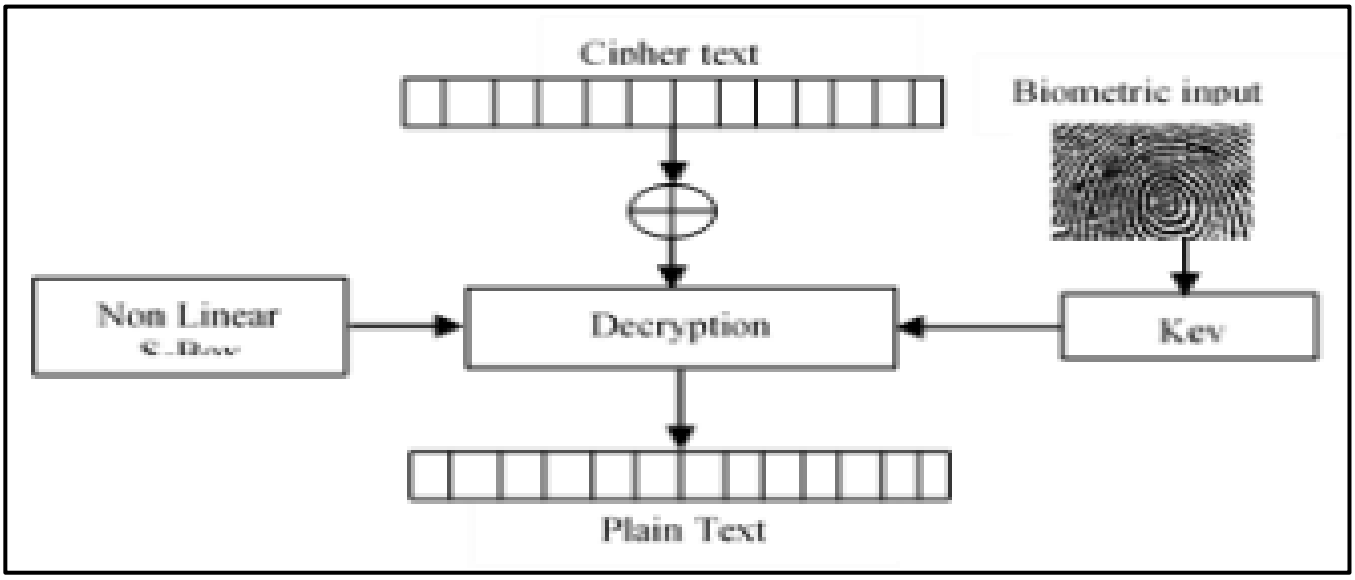
Mix Columns:

Each column of four bytes is now transformed using a special mathematical function. This function takesas input the four bytes of one column and outputs four completely new bytes, which replace the originalcolumn. The result is another new matrix consisting of 16 new bytes. It should be noted that this step is not performed in the last round**.**

Add round key:

The 16 bytes of the matrix are now considered 128 bits and are XORed to the 128 bits of the round key.If this is the last round then the output is the ciphertext. Otherwise, the resulting 128 bits are interpreted as 16 bytes and we begin another similar round.

Decryption Process:



The process of decryption of an AES cipher text is similar to the encryption process in reverse order. Each round consists of the four processes conducted in the reverse order −

- Add round key
- Mix columns
- Shift rows
- Byte substitution

Since sub-processes in each round are in a reverse manner, unlike for a Feistel Cipher, the encryption and decryption algorithms need to be separately implemented, although they are very closely related.

- **Hashing:**

The encrypted value consists of the numbers and letters which don't resemble the original data and is called the hash. Hashing requires processing the data from a block through a mathematical function, which results in an output of a fixed length. Using a fixed-length output increases security since anyone trying to decrypt the hash won't be able to tell how long or short the input is simply by looking at the length of the

output. Solving the hash starts with the data available in the block header and is essentially solving a complex mathematical problem. Each block header contains a version number, a timestamp, the hash used in the previous block, the hash of the Merkle Root, the nonce, and the target hash. The miner focuses on the nonce,a string of numbers. This number is appended to the hashed contents of the previous block, which is then hashed. If this new hash is less than or equal to the target hash, then it is accepted as the solution, the miner is given the reward, and the block is added to the blockchain. Solving the hash requires the miner to determine which string to use as the nonce, which itself requires a significant amount of trial-and-error. This is becausethe nonce is a random string. It is highly unlikely that a miner will successfully come up with the correct nonceon the first try, meaning that the miner may potentially test a large number of nonce options before getting it right. The greater the difficulty—a measure of how hard it is to create a hash that meets the requirement of the target hash—the longer it is likely to take to generate a solution.

- **SHA-256 Algorithm:**

SHA-2 (Secure Hash Algorithm 2) is a set of cryptographic hash functions designed by the United States National Security Agency (NSA) and first published in 2001. They are built using the Merkle–Damgård construction, from a one-way compression function itself built using the Davies–Meyer structure from a specialized block cipher.SHA-2 includes significant changes from its predecessor, SHA-1. The SHA-2 family consists of six hash functions with digests (hash values) that are 224, 256, 384 or 512 bits: SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, SHA-512/256. SHA-256 and SHA-512 are novel hash functions computed with eight 32-bit and 64-bit words, respectively. They use different shift amounts and additive constants, but their structures are otherwise virtually identical, differing only in the number of rounds. SHA-224 and SHA-384 are truncated versions of SHA-256 and SHA-512 respectively, computed with different initial values. SHA-512/224 and SHA-512/256 are also truncated versions of SHA-512, but the initial values are generated using the method described in Federal Information Processing Standards (FIPS) PUB 180-4.SHA-2 was first published by the National Institute of Standards and Technology (NIST) as a U.S. federal standard (FIPS). The SHA-2 family of algorithms are patented in US patent 6829355. The United States has released the patent under a royalty-free license. Currently, the best public attacks break preimage resistance for 52 out of 64 rounds of SHA-256 or 57 out of 80 rounds of SHA-512, and collision resistance for 46 out of 64 rounds of SHA-256.

Finger print recognition system:

Input finger print Binary image Minutiae extraction

Fingerprint is a physical trait of human beings. It is used as a biometric feature and is extracted through biometric processing. Biometric is used here to generate the key, used for data encryption and decryption. Biometric processing includes various operations such as capturing analog data, preprocessing, minutiae extraction and key generation.

Input fingerprint image is initially segmented with an intention of noise removal. The entire image is divided into matrix of size 16x16. Variance is then calculated and is compared with defined global threshold value (0.10). This is accomplished for the entire image. If the value of variance is less than threshold value, the value is deleted. The image is later normalized to increase image quality, by obtaining desired variance.

The image is binarized using Fixed Thresholding Binarization method which takes an image and returns a binary value. In this method fixed threshold value is used to assign 0's and 1's for all pixel positions. It does so by using two thresholds, one for background and one for fingerprint. The image will undergo padding with padding number of pixels from every side. Each of these padded pixels will be "painted" in black.

Crossing Number (CN) concept is used for minutiae extraction. By examining neighborhood of each ridge pixel using a 3x3 window. This method extracts ridge endings and bifurcations from the skeleton image. Ridge ending pixel corresponds to a Crossing Number of one and bifurcation pixel corresponds to a Crossing Number of three. Neighborhood of P, of pixel p. Each minutiae points extracted from a fingerprint image is denoted as (x, y) coordinates. In this, we store those extracted minutiae points in two different vectors, Vector M1 comprises every x co-ordinate values and vector M2 comprises every y co-ordinate values. By using M1 and M2 128-bit biometric key is generated.

Introduction to GSM (Global System for Mobile Communications)

The **Global System for Mobile Communications** (**GSM**) is a standard developed by the European Telecommunication Standard (ETSI) to describe the protocols for second-generation (2G) digital cellular networks used by mobile devices such as mobile phones and tablets. It was first deployed in Finland in December 1991. By the mid-2010s, it became a global standard for mobile communications achieving over 90% market share, and operating in over 193 countries and territories.

2G networks developed as a replacement for first generation (1G) analog cellular networks. The GSM standard originally described a digital, circuit-switched network optimized for full duplex voice telephony. This expanded over time to include data communications, first by circuit-switched transport, then by packet data transport via General Packet Radio Service (GPRS), and Enhanced Data Rates for GSM Evolution (EDGE). Subsequently, the 3GPP developed third-generation (3G) UMTS standards, followed by the fourth-generation (4G) LTE Advanced and the fifth-generation 5G standards, which do not form part of the ETSI GSM standard.

1.2 Literature Survey

Biometrics, with its uniqueness to every individual, has been adapted as a security authentication feature by many institutions. These biometric data are processed into templates that are saved on databases, and a central authority centralizes and controls these databases. This form of storing biometric data, or in our case fingerprint template, is asymmetric and prone to three main security attacks, such as fake template input, template modification or deletion, and channel interception by a malicious attacker. In this paper, we secure an encrypted fingerprint template by a symmetric peer-to-peer network and symmetric encryption. The fingerprint is encrypted by the symmetric key algorithm: Advanced Encryption Standard (AES) algorithm and then is uploaded to a symmetrically distributed storage system, the InterPlanetary File system (IPFS). The hash of the templated is stored in a decentralized blockchain. The slow transaction speed of the blockchain has limited its use in real-life applications, such as large file storage, hence, the merge with IPFS to store just the hashes of large files. The encrypted template is uploaded to the IPFS, and its returned digest is stored on the Ethereum network. The implementation of IPFS prevents storing the raw state of the fingerprint template on the Ethereum network in order to reduce cost and also prevent identity theft. This procedure is an improvement of previous systems. By adopting the method of template hashing, the proposed system is cost-effective and efficient. The experimental results depict that the proposed system secures the fingerprint template by encryption, hashing, and decentralization.

1.3 Problem Definition

In recent days we find many fraudulent activities taking place during the examination. In the way of finding arelevant solution for this problem usage of biometrics with blockchain is found to be apt. To achieve this various encryption and decryption algorithms are being used which provides security to the biometric data without the involvement of a third party.

The main aim of this project is to provide a trustworthy way to authenticate biometrics to reduce malpracticesand save time. It also aims at reducing the risk of cyberattacks on biometric information. This level of securityis observed to be possible only with blockchain, considering its security algorithms.

Another level of security can be added using GSM (Global Standard for Mobile communications) telecommunication standard which uses OTP to authenticate the user. To control all these, we require a user interface controlled by the admin which is created using various web technologies like HTML, CSS, JavaScript, PHP, etc.

1.4 Existing System

The biometric authentication process is something that is mostly seen in airport security, law enforcement, mobile access and authentication, banking, home assistants, etc. Its usage is least seen in the field of educationlimited to the attendance system. Exam authentication system uses checking of hall tickets which use no technology and also consumes more time. There are cases where the student uses some other person to get hisexamination done. To restrict these malpractices, one must different technologies like biometric verification. Though it is used in some institutions for exam gate authentication purposes, the way of storing the biometricinformation is using traditional databases. Using databases to some extent is useful but gives a high scope to information modification. This leads to cyber-attacks and the editing of sensitive biometric information. In this situation, any person can attend the examination though he is not an authorized student. In this system, there is the involvement of the third person which reduces the rate of privacy.

1.4 Proposed system

Biometric authentication has become a very essential step for any verification process. But usage of the database for this is not appropriate, as it doesn't provide the required security. But there is blockchain technology that the best replaces database technology. Blockchain technology when compared to the traditional database technology provides the users with trust, privacy, integrity, resiliency. Blockchain provides a decentralized, transparent database that ensures trust between network participants. The participants can be partners in a supply chain or businesses in a consortium, who use blockchain to record transactional data and other shared information. Due to the blockchain's democratization of control, it enables all participants to have equal access to the stored data and seeks consensus from all participants before approving any change. This decentralized trust model can enable the formation of secure ecosystems that groups of companies can use for storing and sharing data. Data integrity is a foundational property of data stored on the blockchain. Accessing and modifying the data stored on the blockchain is nearly impossible without notifying and seeking consensus from the entire network. Thus, blockchain can be used as the sourceof truth by participants and drive a secure ecosystem that can function in a trustless manner, i.e., without requiring any trust or even familiarity with the other party. Blockchain technology uses different security algorithms like hashing to provide the required security. Besides this project also aims at providing 2-factor authentication using OTP. This system is operated using the user interface handled by an admin. It fails whenthe student's biometrics are found to be unmatched for the continuous 3 times. It helps the organizations to run the examination process without any malpractice.

2 SYSTEM REQUIREMENTS

2.1 Hardware Requirements:

The development and deployment of the application require the following general and specific minimum requirements for hardware:

Component	Minimum Requirement
Processor andSpeed	64-bit, four-core, 2.5 GHz minimum per core
RAM capacity	4 GB for development and evaluation use
Hard Disk	10GB for development and evaluation use in a total capacity of 80GB
Scanning Devices	Cogent CSD 200

2.2 Software Requirements:

The development and deployment of the application require the following general and specific minimum requirements for software:

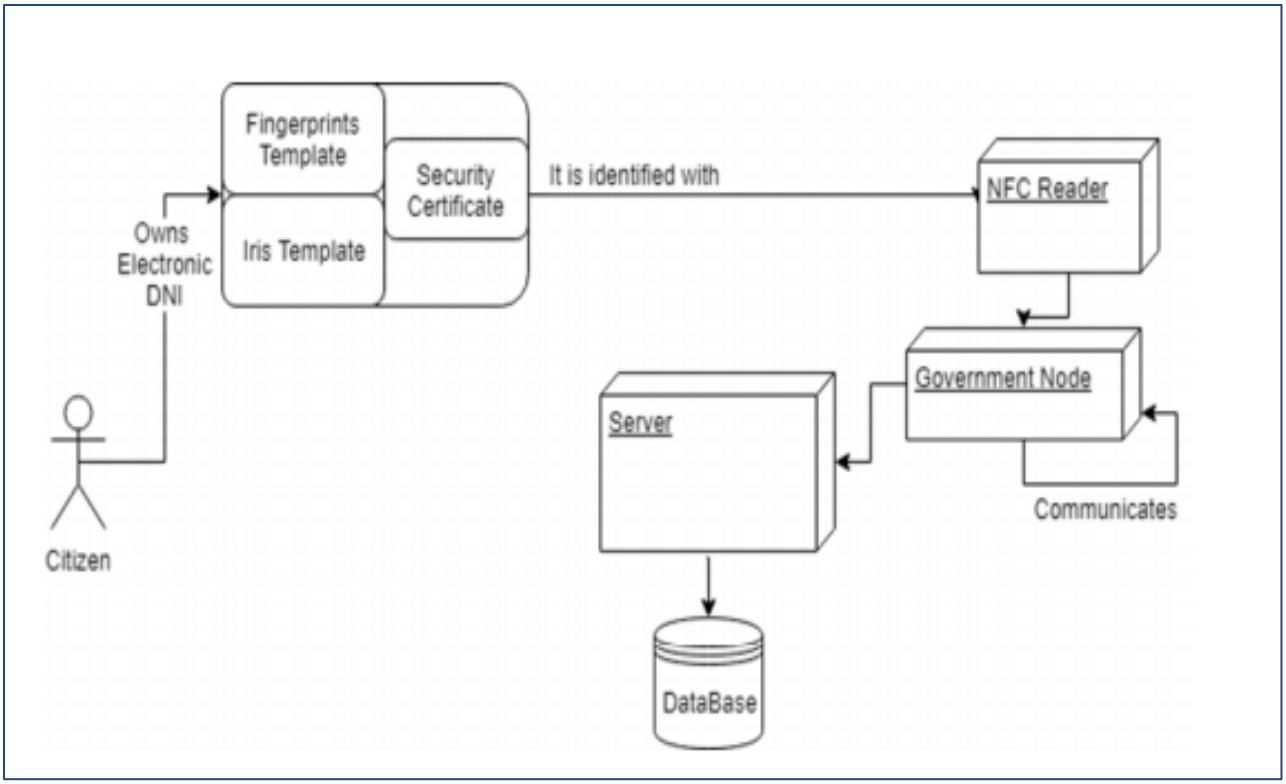
Component	Minimum Requirements
Operating System	Windows Server 2008 R2(64-bit)
Blockchain environment	Ethereum
Web Technologies	HTML, CSS, JavaScript, PHP, Django

3 SYSTEM DESIGN

3.1 Dataflow diagrams / UML diagrams:

Using database:

When the data is collected and stored in the databases there are many threats to the data when compared to a blockchain. Let us consider the below diagram. The user provides the biometrics to the admin which include fingerprints, Iris or Security certificate. It is then recognized using the NFC Reader. The government Node is the electronic device which acts the in-charge to control all the transactions. Later, without any encryption process, the biometrics are stored in the database which is prone to a high risk.



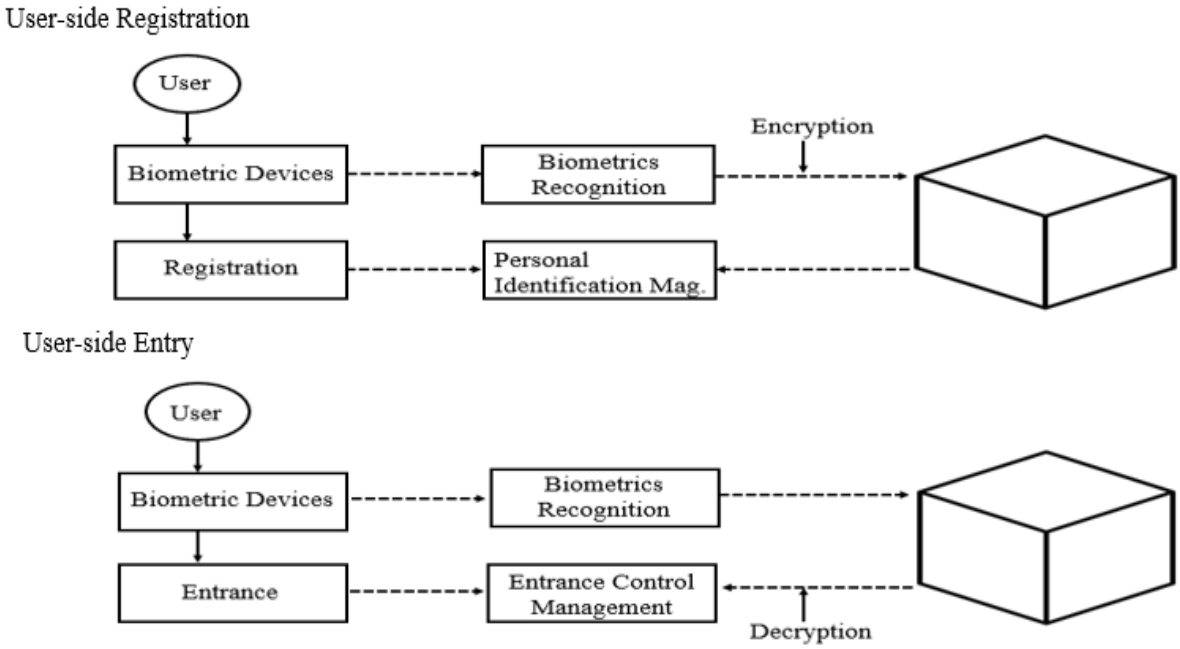
To overcome these risks, we can use the emerging technology, blockchain. In the blockchain, there is no risk for cyber threats because of its high-security standards. The architecture used for blockchain is as follows:

Using Blockchain:

The architecture has two phases. One is the User-side registration phase and the other is the User-side entry phase.

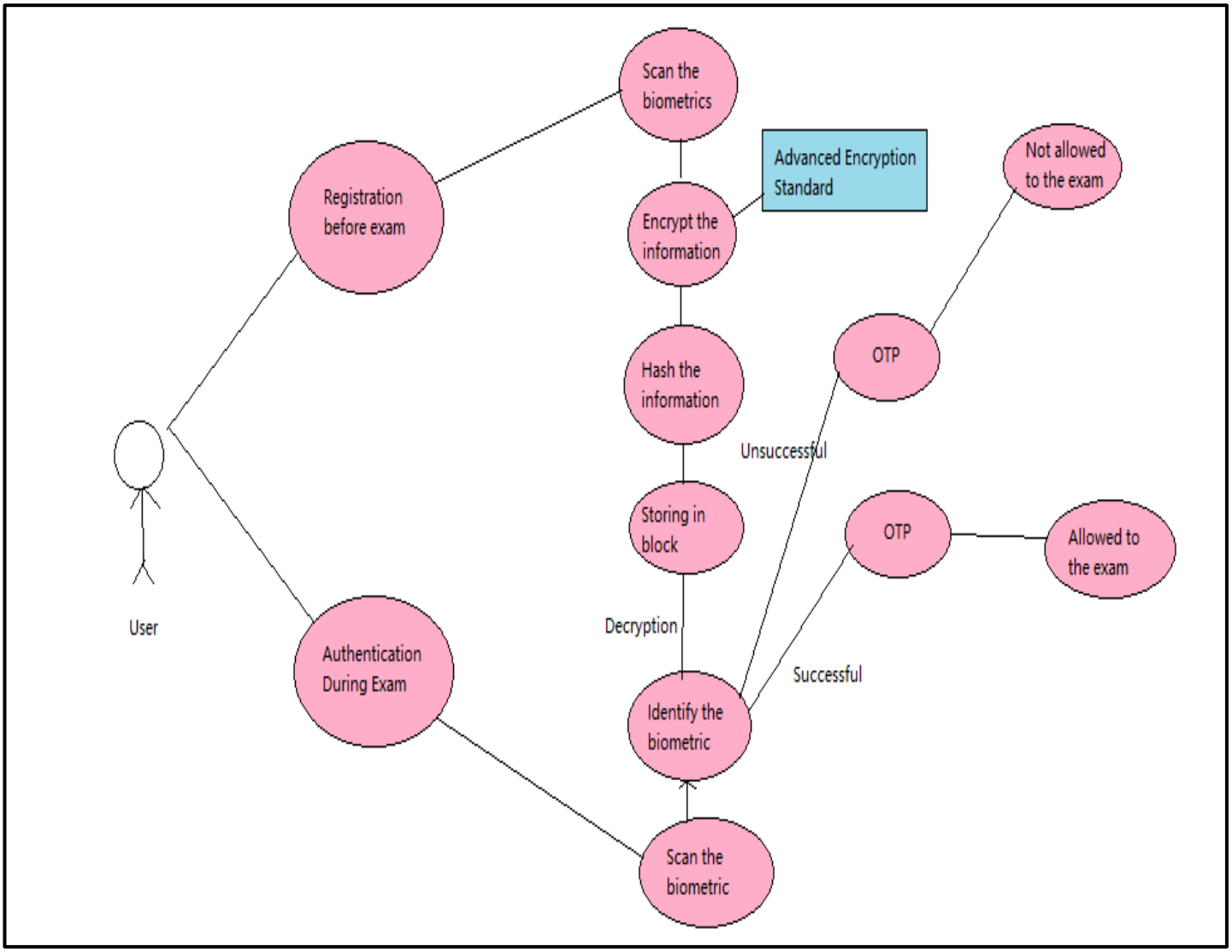
In the User side - Registration phase, there are two kinds of inputs; from the admin and the student. The details are provided to the system through the GUI created according to the admin and the student. These details containing the biometrics are recognized and encrypted using Advanced Encryption Standard and stored in the block which results in the generation of OTP leading to a successful registration process.

In the User side – Entry phase, the admin first logs in through the ID and gets confirmed through an OTP. Later the student enters his/her details and gets the biometric scanned. The scanned biometric gets recognized and automatically matches itself with the biometrics in the blockchain using the decryption process. If the student biometric details are found then the OTP is generated and the student is sent to the examination hall resulting in a successful entry process.



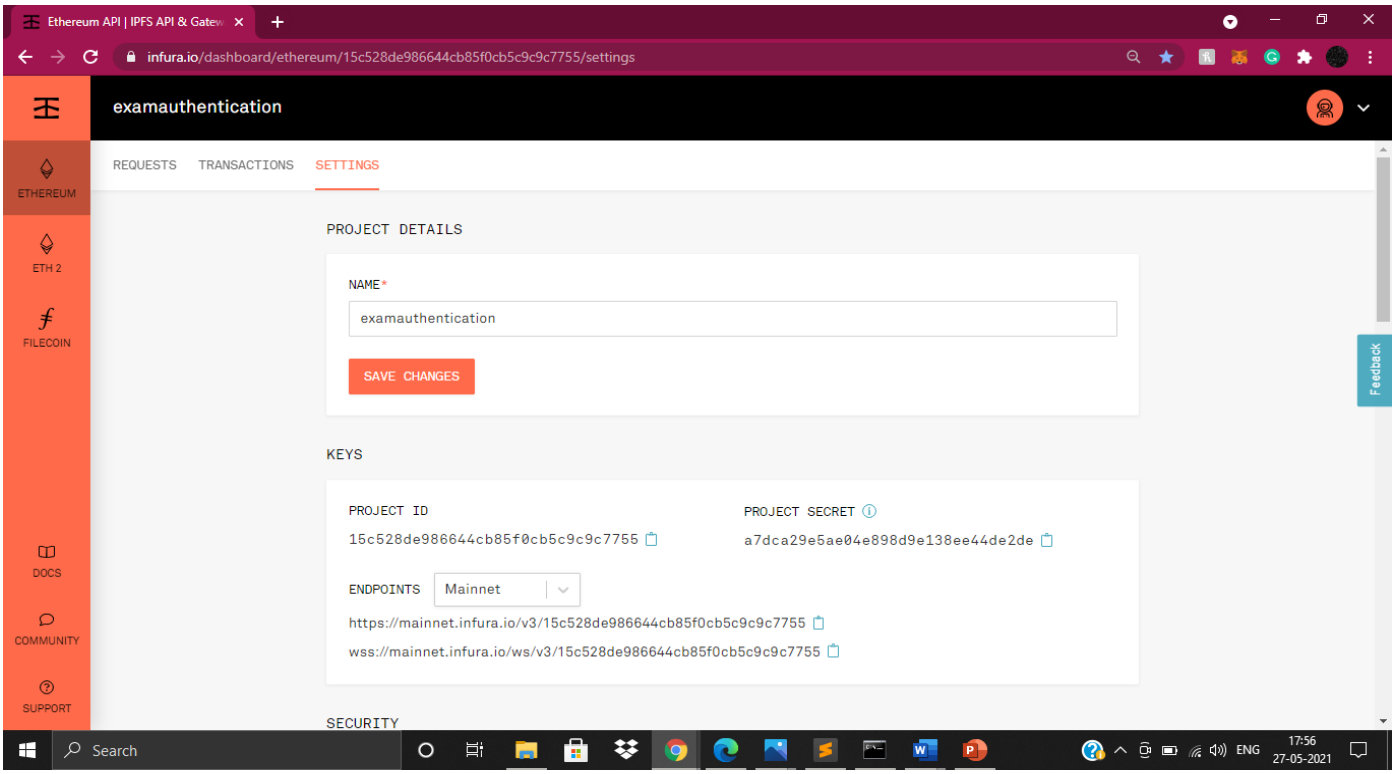
3.2 User case Diagrams:

This diagram depicts the function of the design system clearly. Its also shows the use of the Advanced Encryption Standard in both the phases; the entry phase and the registration phase. This diagram also depicts the OTP generation process which verifies the student as well as the admin.

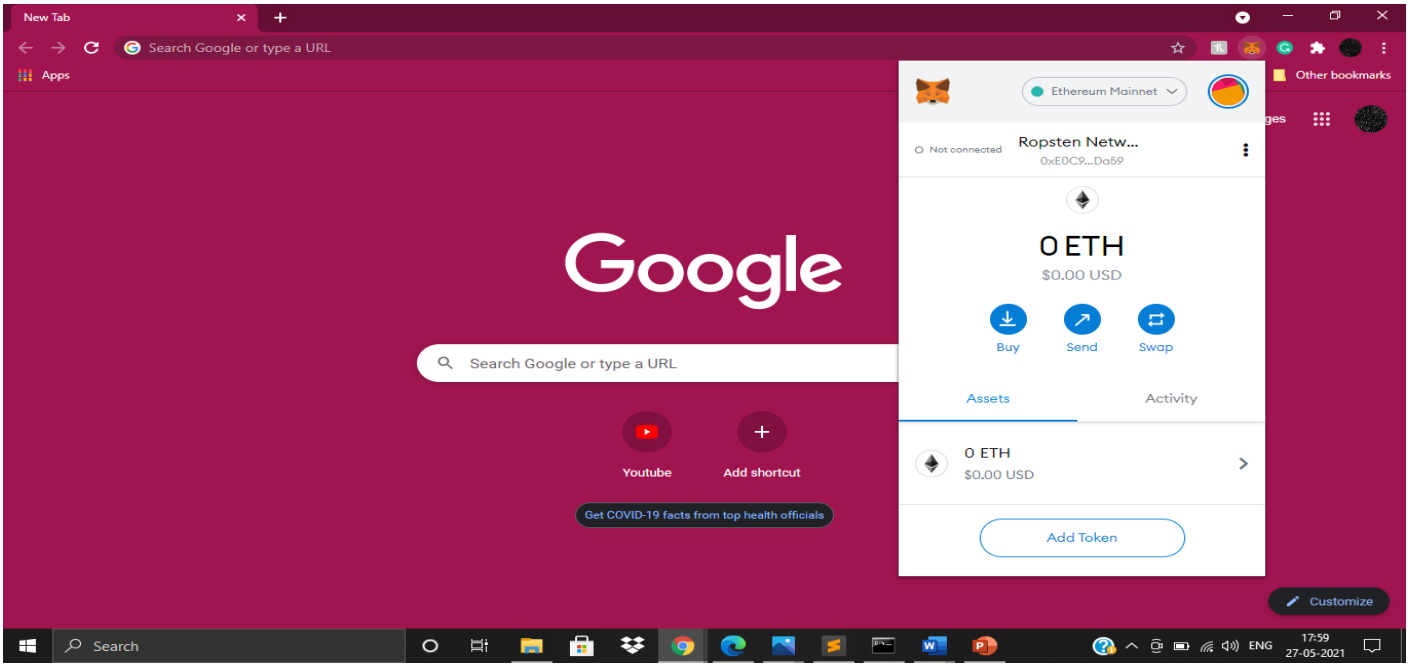


4 IMPLEMENTATION

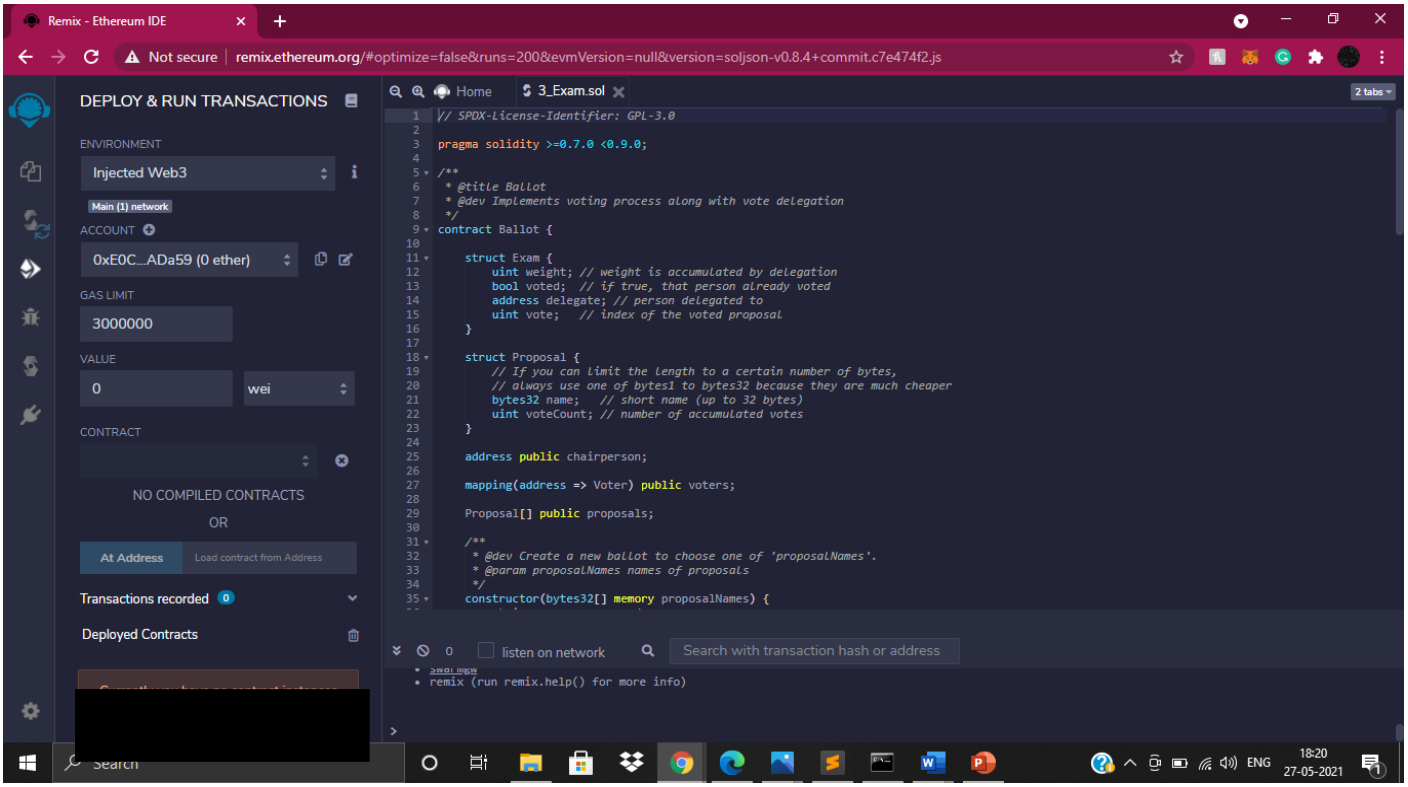
Step -1 Get the Product key using infura.io



Step-2 Install Meta-mask addon



Step-3 Connect web3 and the sublime text using remix IDE and solidity code (Block chain connected to Sublime text)



Step-4 Install web3

```
C:\Users\KESAVA REDDY>pip install web3
```

Step-5 Connect to web3.py and initiate the interaction

```
import json
from web3 import web3
from .render import render
from django.http import HttpResponseRedirect
from django.contrib import messages
from django.core.paginator import paginator
from django.shortcuts import render,redirect,HttpResponseRedirect
from django.contrib.auth.decorators import login_required
```

```
public_key="15c528de986644cb85f0cb5c9c9c7755"
private_key="a7dca29e5ae04e898d9e138ee44de2de"
url='https://ropsten.infura.io/v3/15c528de986644cb85f0cb5c9c9c7755'
web3=web3(Web3.HTTPProvider(url))
address=web3.toCheckSumAddress("15c528de986644cb85f0cb5c9c9c7755")
abi=json.loads('''[]''')
contract=web3.eth.contract(address=address,abi=abi)
```

Step-6 Install Django

```
C:\Users\KESAVA REDDY>pip install django
```

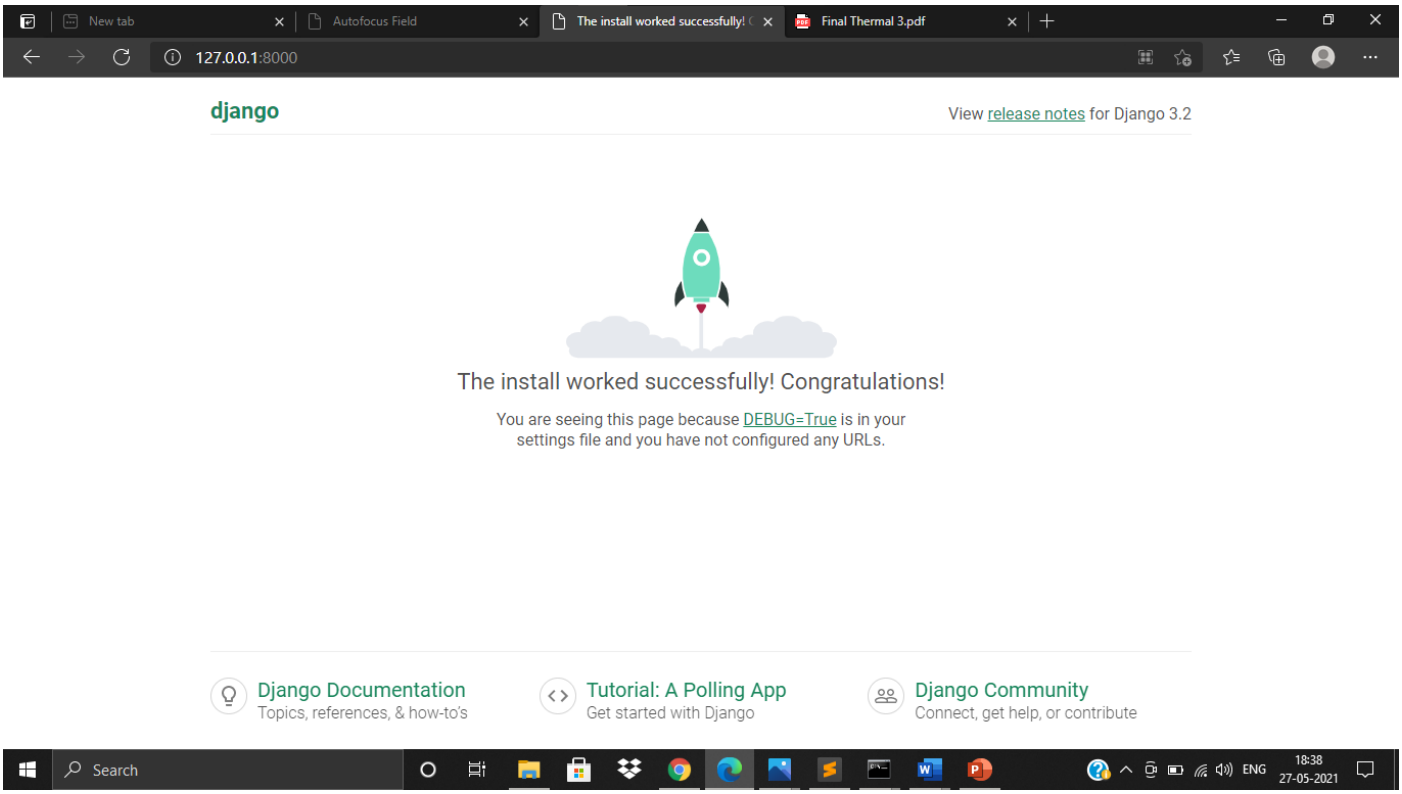
Step-7 Importing django

```
from django.urls import path
from .views import SignUpView
urlpatterns = [
    path('signup/', SignUpView.as_view(), name='signup'),
]
```

Step- 8 Run local server

```
Command Prompt - python manage.py runserver

C:\Users\KESAVA REDDY\Desktop>cd accounts

C:\Users\KESAVA REDDY\Desktop\accounts>django-admin.py startproject examreg
C:\Users\KESAVA REDDY\AppData\Local\Programs\Python\Python39\Scripts\django-admin.py:17: RemovedInDjango40Warning: django-admin.py is deprecated in favor of django-admi
n.
  warnings.warn(

C:\Users\KESAVA REDDY\Desktop\accounts>cd examreg

C:\Users\KESAVA REDDY\Desktop\accounts\examreg>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
May 27, 2021 - 14:44:43
Django version 3.2.3, using settings 'examreg.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
[27/May/2021 14:45:00] "GET / HTTP/1.1" 200 10697
[27/May/2021 14:45:00] "GET /static/admin/css/fonts.css HTTP/1.1" 304 0
[27/May/2021 14:45:01] "GET /static/admin/fonts/Roboto-Light-webfont.woff HTTP/1.1" 304 0
[27/May/2021 14:45:01] "GET /static/admin/fonts/Roboto-Regular-webfont.woff HTTP/1.1" 304 0
[27/May/2021 14:45:01] "GET /static/admin/fonts/Roboto-Bold-webfont.woff HTTP/1.1" 304 0
Not Found: /favicon.ico
[27/May/2021 14:45:01] "GET /favicon.ico HTTP/1.1" 404 2111
[27/May/2021 16:11:02] "GET / HTTP/1.1" 200 10697
```

Step-9 Type http://127.0.0.1:8000/ in the browser. It should display the content in the below diagram which represents the successful set up of server.



Step- 10 Create admin registration files

File Name: Adminreg.html

```
<!DOCTYPE html>
<html>
<head>
        <title>Admin Registration</title>
        <link rel="stylesheet" href="style.css" type="text/css">
</head>
<body>
        <div class="main">
                <div class="register">
                        <h2>RESTRICTED TO ADMIN ONLY</h2>
                        <form id="register" method="post">
                                <label>First Name : </label>
                                <br>
                                <input type="text" name="fname" id="name" placeholder="First Name">
```

```
                    <br><br>
                    <label>Last Name : </label>
                    <br>
                    <input type="text" name="lname" id="name" placeholder="Last Name">
                    <br><br>
                    <label>Employee ID</label>
                    <br>
                    <input type="text" name="eid" id="name" placeholder="Employee ID">
                    <br><br>
                    <label>Phone number : </label>
                    <br>
                    <input type="number" name="mobile" id="name" placeholder="Phone">
                    <br><br>
                    <label>Gender : </label>
                    <br>
                         
                    <input type="radio" name="gender" id="male">
                     
                    <span id="male">Male</span>
                           
                    <input type="radio" name="gender" id="female">
                     
                   <span id="female">Female</span>
                    <br><br>
                    <a href="adminotp.html">
                    <input type="submit" value="submit" name="submit" id="submit">
                    </a>
                </form>
            </div>
        </div>

</body>
</html>
```

File Name: Studentregister.html

```
<!DOCTYPE html>
<html>
<head>
	<title>Student Registration</title>
	<link rel="stylesheet" href="style.css" type="text/css">
</head>
<body>
	<div class="main">
		<div class="register">
			<h2>RESTRICTED TO STUDENTS ONLY</h2>
			<form id="register" method="post">
				<label>Full Name : </label>
				<br>
				<input type="text" name="fname" id="name" placeholder="First Name">
				<br><br>
				<label>Father Name</label>
				<br>
				<input type="text" name="eid" id="name" placeholder="Father Name">
				<br><br>
				<label>Phone number : </label>
				<br>
				<input type="number" name="mobile" id="name" placeholder="Phone">
				<br><br>
				<label>College Name : </label>
				<br>
				<input type="text" name="clgname" id="name" placeholder="College Name”
				<br><br>
				<label>Year: </label>
				<br>
				<input type="text" name="year" id="name" placeholder="Year">
				<br><br>
				<label>Gender : </label>
				<br>
```

```
                         
                    <input type="radio" name="gender" id="male">
                     
                    <span id="male">Male</span>
                           
                    <input type="radio" name="gender" id="female">
                     
                    <span id="female">Female</span>
                    <br><br>
                    <a href="adminotp.html">
                    <input type="submit" value="submit" name="submit" id="submit">
                </a>
            </form>
        </div>
    </div>
</body>
</html>
```

File Name: Style.css

```
*{
    margin: 0;
    padding: 0;
    align-content:center;
}
body{
    background: url("bg.jpg");
    background-size: 100%;
    background-position: center;
}
div.main{

    position: relative;
    left:300px;
    width: 400px;
```

```
  margin: 100px auto 50px auto;
}
h2{
        text-align: center;
        padding: 20px;
        font-family: sans-serif;
}
div.register{
        background-color: rgba(0,0,0,0.5);
        width: 100%
font-size: 18px;
        border-radius: 10px;
        border: 1px solid rgba(255,255,255,0.3);
        box-shadow:2px 2px 15px rgba(0,0,0,0.3);
        color: #fff;
}
form#register{
        margin:40px;
}
label{
        font-family: sans-serif;
        font-size: 18px;
        font-style: "bold";
}
input#name{
        width: 300px;
        border: 1px solid #ddd;
        border-radius: 3px;
        outline: 0;
        padding: 7px;
        background-color: #fff;
        box-shadow: inset 1px 1px 5px rgba(0,0,0,0.3);
}
input#submit{
```

```
        width: 200px;
        padding: 7px;
        font-size: 16px;
        font-family: sans-serif;
        font-weight: 600;
        border: none;
        border-radius: 3px;
        background-color: rgba(250,100,0,0.8);
        color: #fff;
        cursor: pointer;
        border: 1px solid rgba(255,255,255,0.3);
        box-shadow: 1px 1px 5px rgba(0,0,0,0.3);
        margin-bottom: 20px;
}
label,span,h2{
        text-shadow: 1px 1px 5px rgba(0,0,0,0.3);
}
```

File Name: Biometric.html

```
<!DOCTYPE html>
<html>
<head>
<title>Admin Registration</title>
        <link rel="stylesheet" href="biostyle.css" type="text/css">
</head>
<body>
        <div class="main">
                <div class="register">
                        <h2>Place Your Finger On The Scanner</h2>
                        <form id="register" method="post">
                                <label>Finger Print-1</label>
                                <br>
                                <input type="text"  name="fp1" id="name">
                                <br><br>
```

```
                    <label>Finger Print-2</label>
                    <br>
                    <input type="text" rows="25" cols="40" name="fp2" id="name" placeholder>
                    <br><br>
                    <a href="Adminsuccess.html">
                    <input type="submit" value="submit" name="submit" id="submit">
                </a>
            </form>
        </div>
    </div>
</body>
</html>
```

File Name: biostyles.css

```
*{
    margin: 0;
    padding: 0;
    align-content:center;
}
body{
    background: url("bg.jpg");
    background-size: 100%;
    background-position: center;
}
div.main{
     position: relative;
    left:300px;
    width: 400px;
  margin: 100px auto 50px auto;
}
h2{
    text-align: center;
    padding: 20px;
    font-family: sans-serif;
```

```
}
div.register{
        background-color: rgba(0,0,0,0.5);
        width: 100%
        font-size: 18px;
        border-radius: 10px;
        border: 1px solid rgba(255,255,255,0.3);
        box-shadow:2px 2px 15px rgba(0,0,0,0.3);
        color: #fff;
}
form#register{
        margin:40px;
}
label{
        font-family: sans-serif;
        font-size: 18px;
        font-style: "bold";
}
input#name{
  width: 300px;
  height: 150px;
}
input#submit{
        width: 200px;
        padding: 7px;
        font-size: 16px;
        font-family: sans-serif;
        font-weight: 600;
        border: none;
        border-radius: 3px;
        background-color: rgba(250,100,0,0.8);
        color: #fff;
        cursor: pointer;
        border: 1px solid rgba(255,255,255,0.3);
```

```
	box-shadow: 1px 1px 5px rgba(0,0,0,0.3);
	margin-bottom: 20px;
}
label,span,h2{
	text-shadow: 1px 1px 5px rgba(0,0,0,0.3);
}
```

File Name: Success.html

```
<html>
<head>
	<title>Admin Registration</title>
	<link rel="stylesheet" href="successstyle.css" type="text/css">
</head>
<body background="bg.jpg">
	<h2> Registration Successful </h2>
</body>
</html>
```

File Name: unsuccessful.html

```
<html>
<head>
	<title>Admin Registration</title>
	<link rel="stylesheet" href="successstyle.css" type="text/css">
</head>
<body background="bg.jpg">
	<h2> Registration Successful </h2>
</body>
</html>
```

File Name: successstyle.css

```
body{
	background: url("bg.jpg");
	background-size: 100%;
	background-position: center;
```

```
}
h2{
        font-size: 50px;
        font-family: sans-serif;
        text-align: center;
        padding: 20px;
        font-color: white;
}
```

File Name: Adminlogin.html

```
<!DOCTYPE html>
<html>
<head>
        <title>Admin Login</title>
        <link rel="stylesheet" href="style.css" type="text/css">
</head>
<body>
        <div class="main">
                <div class="register">
                        <h2>RESTRICTED TO ADMIN ONLY</h2>
                        <form id="register" method="post">
                                <label>Full Name : </label>
                                <br>
                                <input type="text" name="fname" id="name" placeholder="Full Name”>
                                <br><br>
                                <label>Employee ID</label>
                                <br>
                                <input type="text" name="lname" id="name" placeholder="Employee Id">
                                <br><br>
                                <a href="adminotp.html">
                                <input type="submit" value="submit" name="submit" id="submit">
                          </a>
                        </form>
                </div>
```

```
        </div>
</body>
</html>
```

File Name : Studentlogin.html

```
<!DOCTYPE html>
<html>
<head>
        <title>Student Login</title>
        <link rel="stylesheet" href="style.css" type="text/css">
</head>
<body>
        <div class="main">
                <div class="register">
                        <h2>RESTRICTED TO STUDENTS ONLY</h2>
                        <form id="register" method="post">
                                <label>Full Name : </label>
                                <br>
                                <input type="text" name="fname" id="name" placeholder="Full Name
                                Name”>
                                <br><br>
                                <label>Hall Ticket Number</label>
                                <br>
                                <input type="text" name="lname" id="name" placeholder="Hall Ticket
                                Number”>
                                <br><br>
                                <a href="adminotp.html">
                                <input type="submit" value="submit" name="submit" id="submit">
                          </a>
                        </form>
                </div>
        </div>
</body>
</html>
```

Python Code to scan biometrics using AES

File Name: blck.py

```
# Python programm to create Blockchain
# For timestamp
import datetime
# Calculating the hash
# in order to add digital
# fingerprints to the blocks
import hashlib
# To store data
# in our blockchain
import json
# Flask is for creating the web
# app and jsonify is for
# displaying the blockchain
from flask import Flask, jsonify
class Blockchain:
	# This function is created
	# to create the very first
	# block and set it's hash to "0"
	def __init__(self):
		self.chain = []
		self.create_block(proof=1, previous_hash='0')
	# This function is created
	# to add further blocks
	# into the chain
	def create_block(self, proof, previous_hash):
		block = {'index': len(self.chain) + 1,
				'timestamp': str(datetime.datetime.now()),
				'proof': proof,
				'previous_hash': previous_hash}
		self.chain.append(block)
		return block
```

```
# This function is created
# to display the previous block
def print_previous_block(self):
        return self.chain[-1]
# This is the function for proof of work
# and used to successfully mine the block
def proof_of_work(self, previous_proof):
        new_proof = 1
        check_proof = False
        while check_proof is False:
                hash_operation = hashlib.sha256(
                        str(new_proof**2 - previous_proof**2).encode()).hexdigest()
                if hash_operation[:4] == '00000':
                        check_proof = True
                else:
                        new_proof += 1
                                return new_proof
def hash(self, block):
        encoded_block = json.dumps(block, sort_keys=True).encode()
        return hashlib.sha256(encoded_block).hexdigest()
def chain_valid(self, chain):
        previous_block = chain[0]
        block_index = 1
        while block_index < len(chain):
                block = chain[block_index]
                if block['previous_hash'] != self.hash(previous_block):
                        return False
                previous_proof = previous_block['proof']
                proof = block['proof']
                hash_operation = hashlib.sha256(
                        str(proof**2 - previous_proof**2).encode()).hexdigest()
                if hash_operation[:4] != '00000':
                        return False
                previous_block = block
```

```
                block_index += 1
        return True
# Creating the Web
# App using flask
app = Flask(__name__)
# Create the object
# of the class blockchain
blockchain = Blockchain()
# Mining a new block
@app.route('/mine_block', methods=['GET'])
def mine_block():
    previous_block = blockchain.print_previous_block()
    previous_proof = previous_block['proof']
    proof = blockchain.proof_of_work(previous_proof)
    previous_hash = blockchain.hash(previous_block)
    block = blockchain.create_block(proof, previous_hash)
    response = {'message': 'A block is MINED',
                'index': block['index'],
                'timestamp': block['timestamp'],
                'proof': block['proof'],
                'previous_hash': block['previous_hash']}
    return jsonify(response), 200
# Display blockchain in json format
@app.route('/get_chain', methods=['GET'])
def display_chain():
    response = {'chain': blockchain.chain,
                'length': len(blockchain.chain)}
    return jsonify(response), 200
# Check validity of blockchain
@app.route('/valid', methods=['GET'])
def valid():
    valid = blockchain.chain_valid(blockchain.chain)
    if valid:
        response = {'message': 'The Blockchain is valid.'}
```

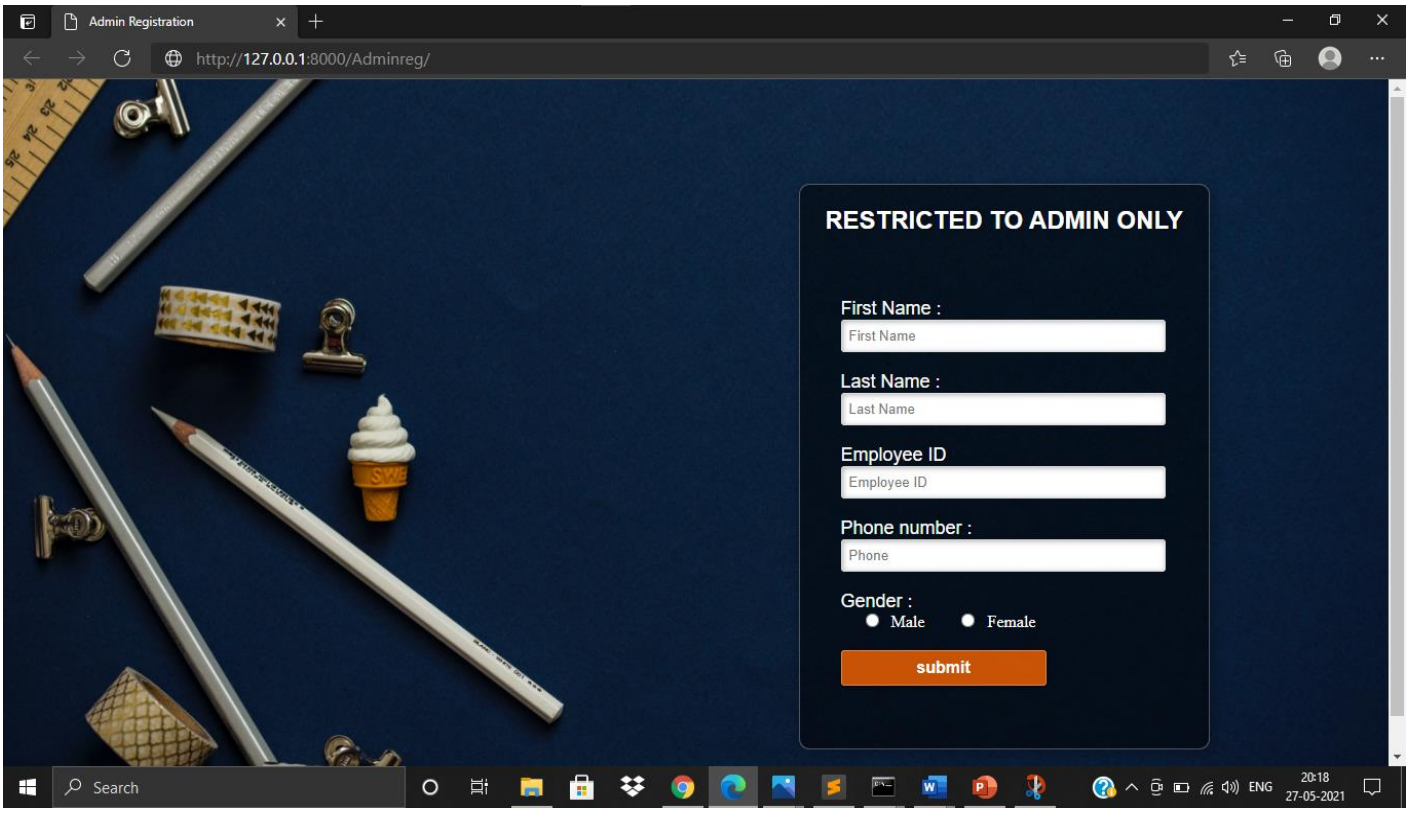
```
        else:
                response = {'message': 'The Blockchain is not valid.'}
        return jsonify(response), 200
# Run the flask server locally
app.run(host='127.0.0.1', port=5000)
```

Python file to encrypt the ipfs file:

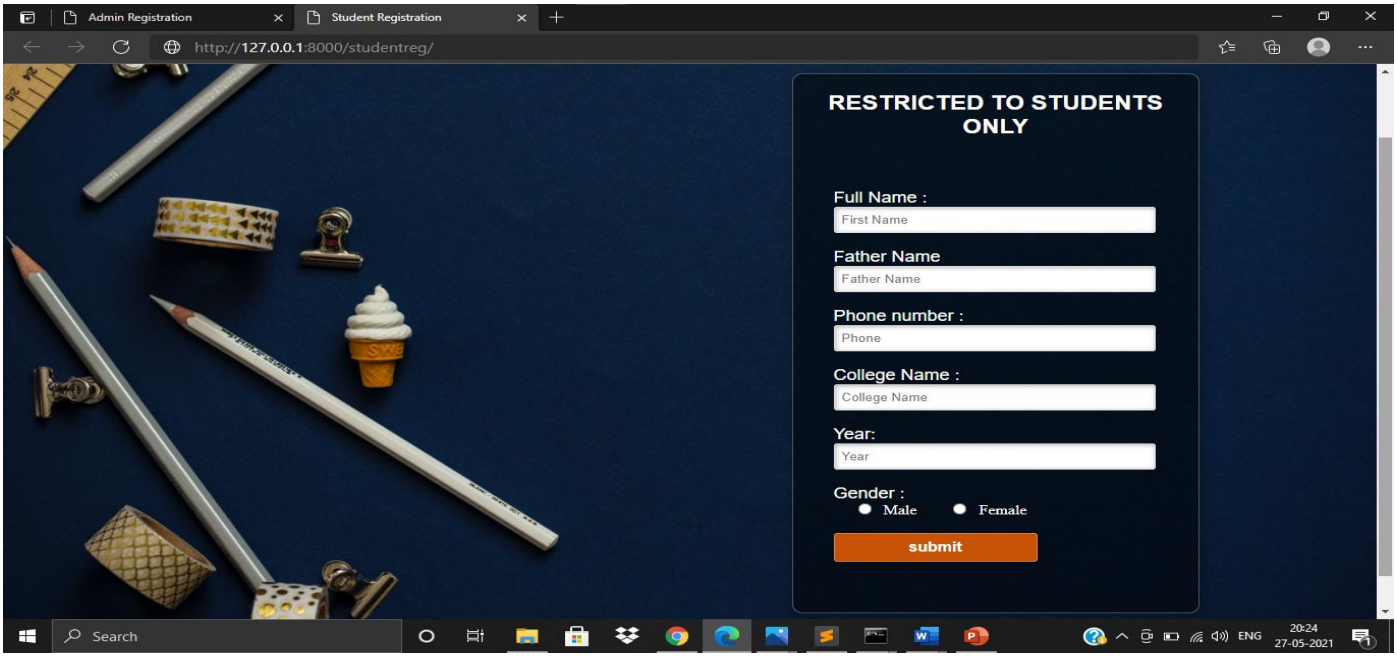
```
import rsa
 # generate public and private keys with
# rsa.newkeys method,this method accepts
# key length as its parameter
# key length should be atleast 16
publicKey, privateKey = rsa.newkeys(512)
# this is the string that we will be encrypting
message = "QmUf3HUg5RoQ43pTKbRHw7fTDxyioZUNa4wjn29ZtuFMsN"
# rsa.encrypt method is used to encrypt
# string with public key string should be
# encode to byte string before encryption
# with encode method
encMessage = rsa.encrypt(message.encode(), publicKey)
print("original string: ", message)
print("encrypted string: ", encMessage)
# the encrypted message can be decrypted
# with ras.decrypt method and private key
# decrypt method returns encoded byte string,
# use decode method to convert it to string
# public key cannot be used for decryption
decMessage = rsa.decrypt(encMessage, privateKey).decode()
print("decrypted string: ", decMessage)
```

5 RESULTS

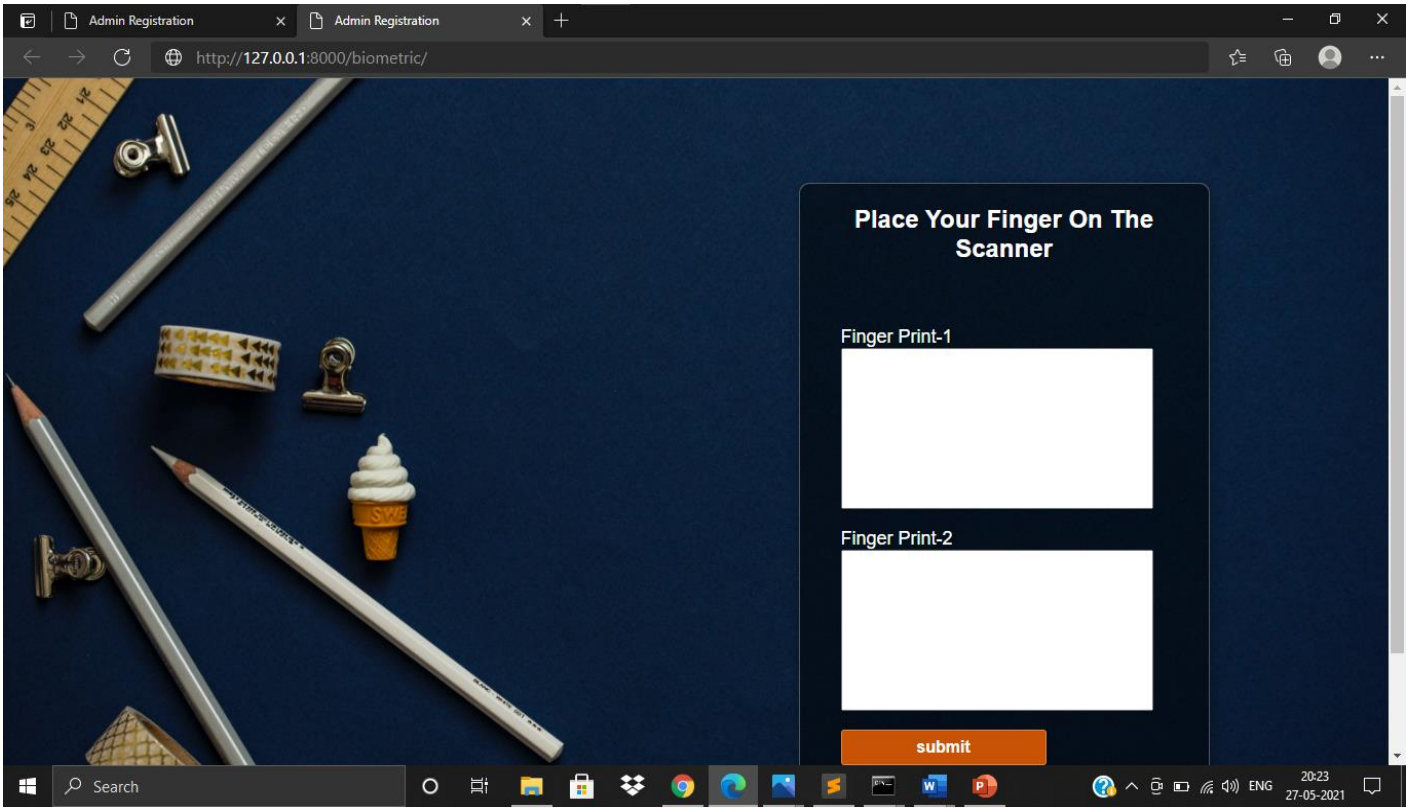
File Name: Adminreg.html



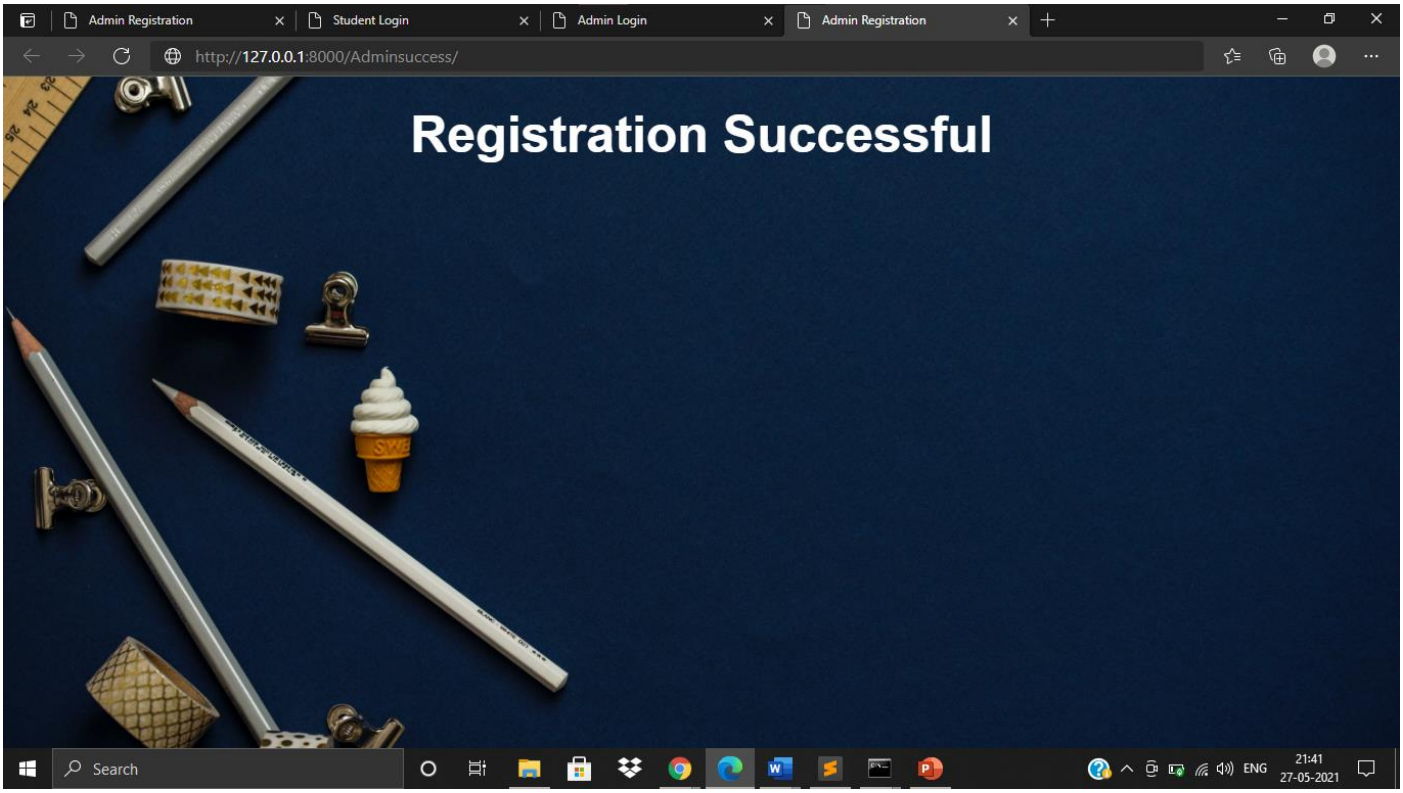
File Name: Studentregister.html



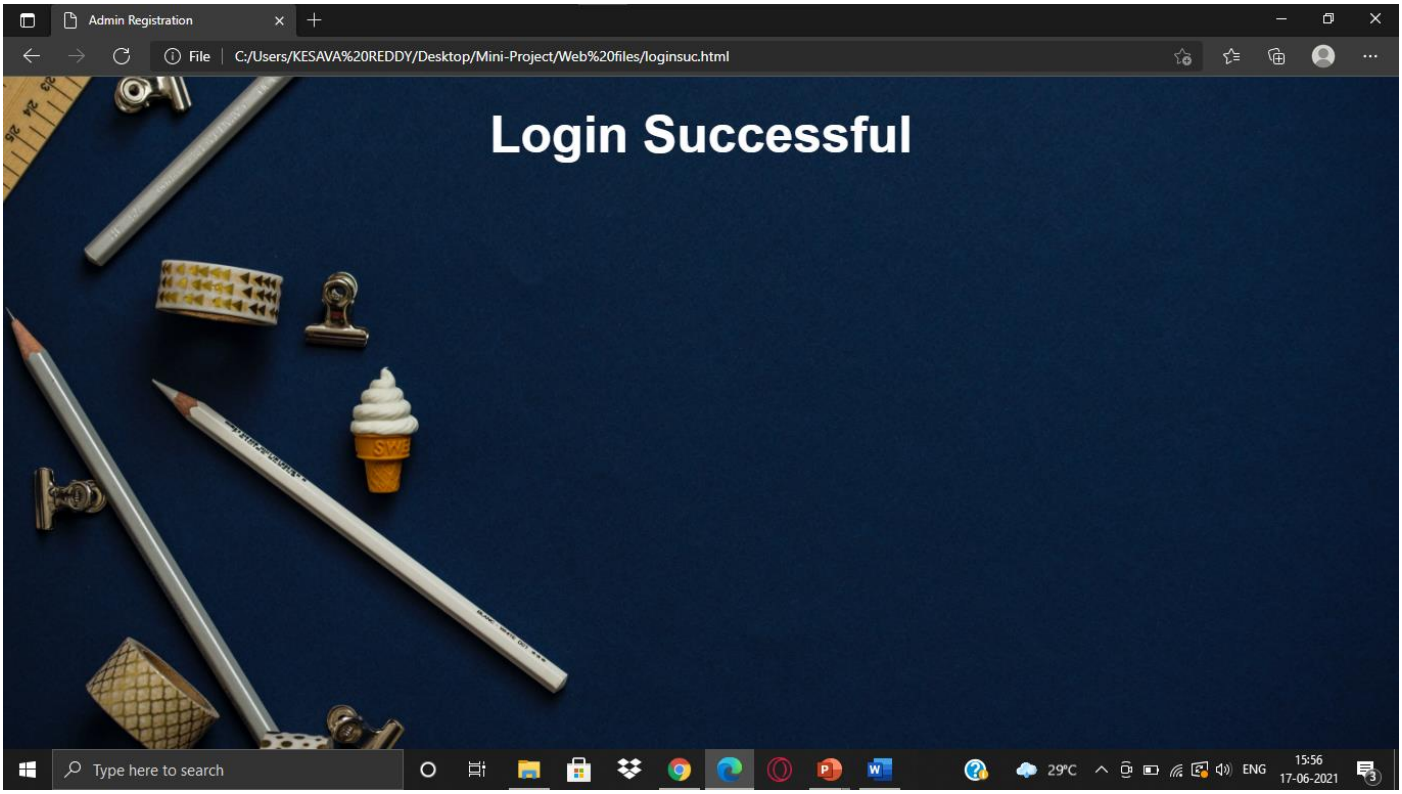
File Name: Biometric.html



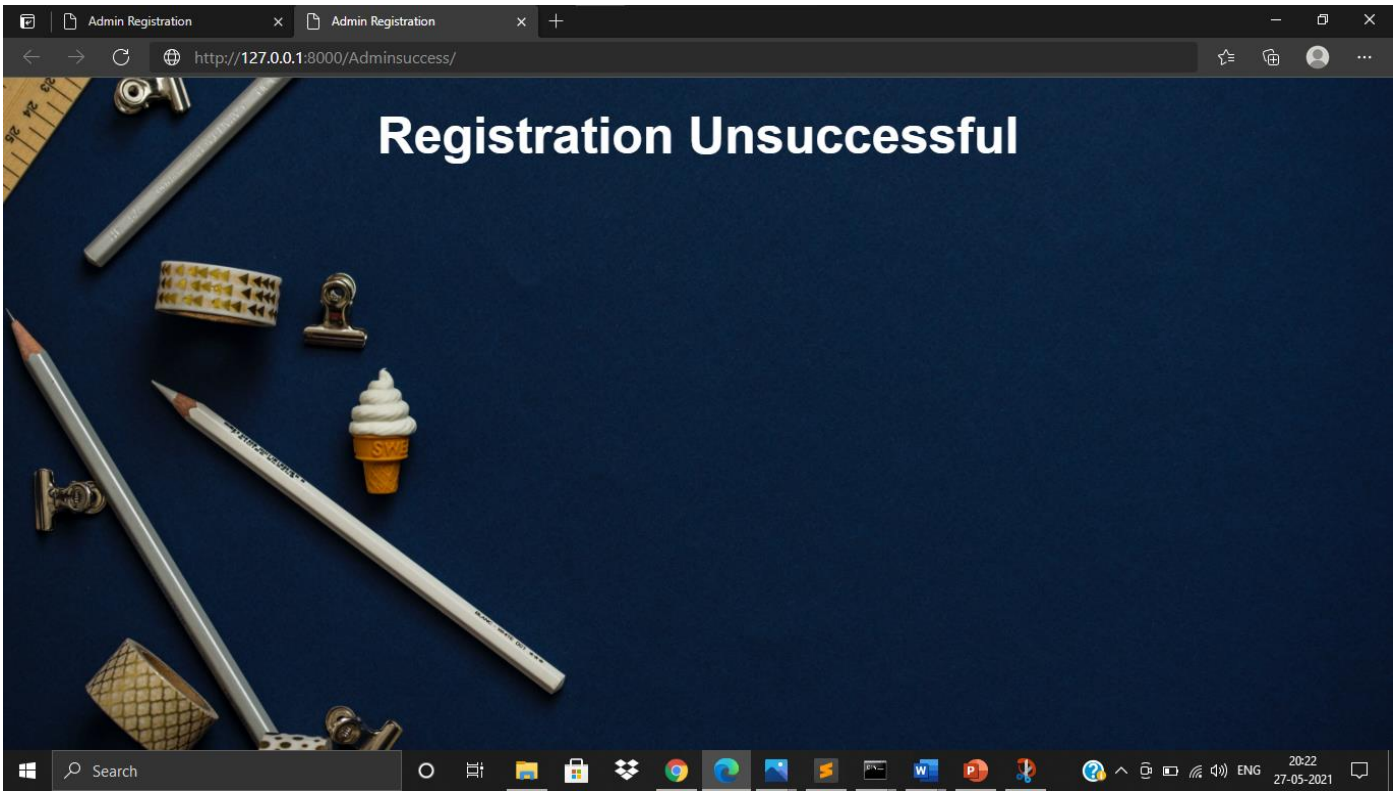
File Name: Success.html



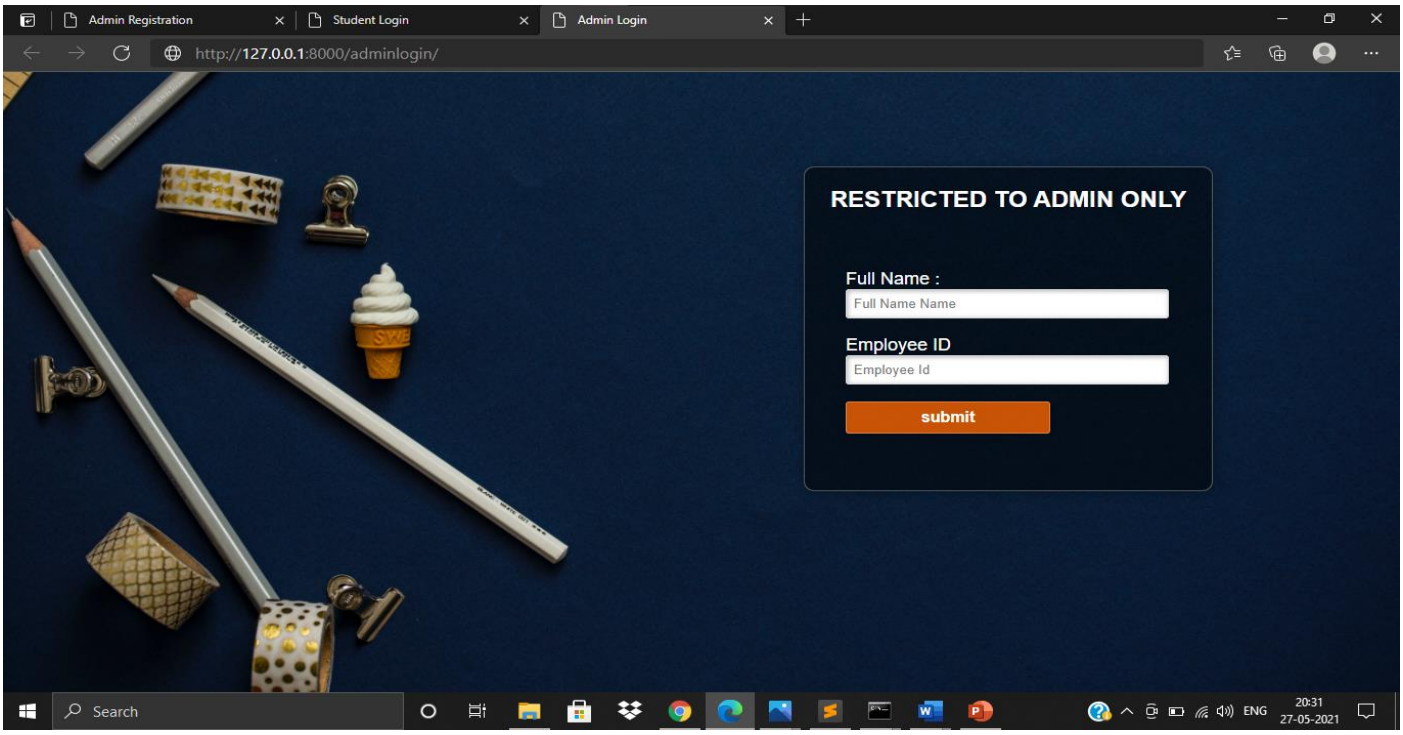
Loginsuccess.html:



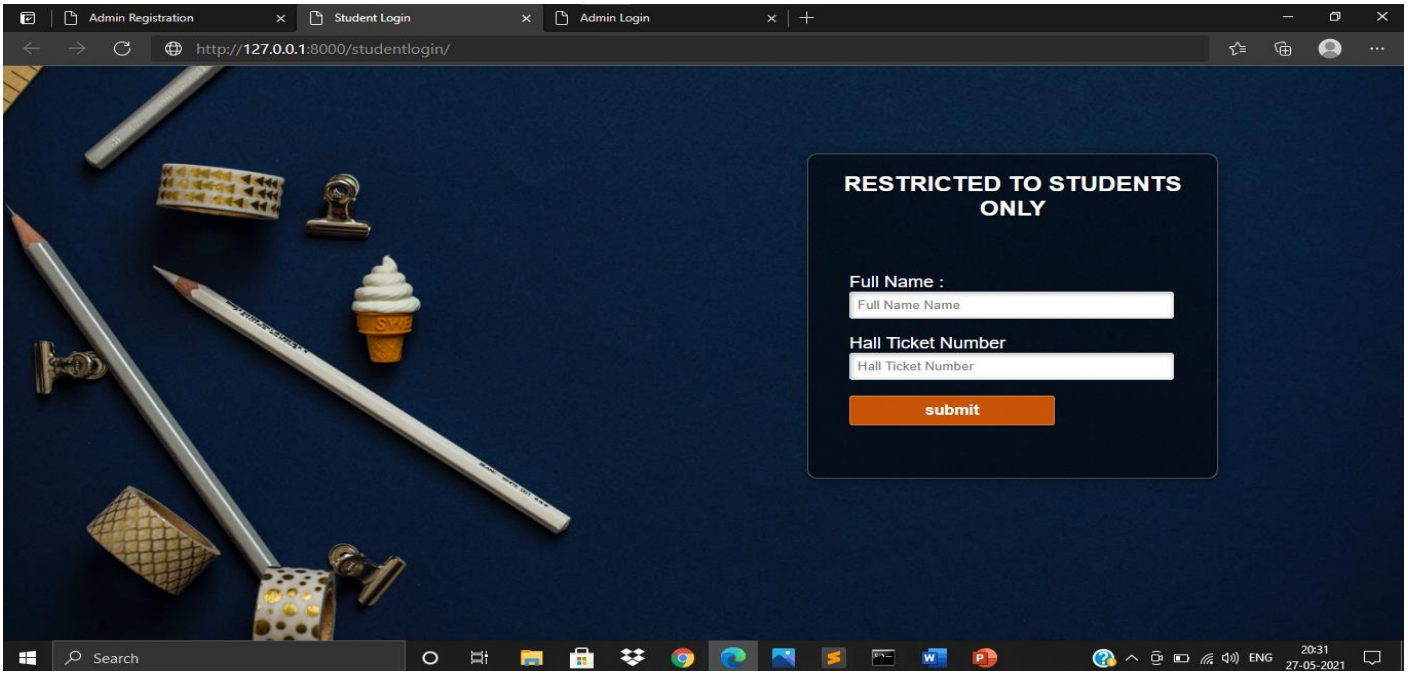
File Name: unsuccessful.html



File Name: Adminlogin.html



File Name: Studentlogin.html



Encryption and Decryption:

```
original string:  QmUf3HUg5RoQ43pTKbRHw7fTDxyioZUNa4wjn29ZtuFMsN
encrypted string:  b'd\xa4\xa9\xa4DI3;o\xf0\xb5\x94VHN\xe3\x16\xb8\xa8\x86\xcec\xf4\xf0\xab\xa4\xe0\x13`\xf7\x8bK\xba\xe5B7\xc6
\xb5q\x99\xa9r\xa6\xe7\xa9B\xf0c\xf16\xbf\xb7\xfa\xd3\t\xf5\xad\xfd\x85\x80\x88:E\xfd'
decrypted string:  QmUf3HUg5RoQ43pTKbRHw7fTDxyioZUNa4wjn29ZtuFMsN
```

6 CONCLUSION

In this mini-project, we learned to use the blockchain to store biometrics. We also learned about many technologies like blockchain, GSM, Ethereum, and web technologies.

We used Advanced Encryption Standard (AES) algorithm to encrypt and decrypt the biometric input given. We also created a GUI using various web technologies for the admin to interact with the server and retrieve or register the biometrics. We used GSM technologies to generate OTP for the successful authentication of the student and admin.

Using blockchain means providing high security to the data inserted into the blockchain. Blockchain itself provides two steps of encryption; AES and Hashing. When it also meets OTP authentication, the system will be at its zenith. Blockchain also doesn't allow its users to change the data in the blocks so easily, because, data in every block is linked to the hash and every hash is linked to the hash of the next block, and so on. This makes it difficult for the user to change the data as it needs to change the hash of all the blocks. This also provides security from the hackers as they cannot go through every hash at a time.

When three security standards; biometrics, blockchain, and OTP integrate there is no place for fraudulent activities to take place.

Future Scope

Blockchain is an emerging technology that makes it difficult to use it with biometrics and GSM. But when these three technologies integrate, this system will be at zenith. In this paper, Ethereum and Remix IDE are used as the platform for blockchain. Though Ethereum is a standard environment for blockchain its usage is difficult for beginners. Usage of IBM blockchain can also be a great option. Secondly, JavaScript is used for the generation of OTP which neglects the high usage of GSM. Finally, for the GUI, Django, HTML, CSS, and JavaScript are being used. This report tries to integrate GSM and different kinds of biometrics to benefit every individual. This report also tries to integrate methods that provide security for the admin regarding the submission of eligibility documents beforehand. This idea integrates all the above-mentioned loopholes in the future for greater accuracy and security.

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